Name:

Quiz 1 Solutions

Instructions. You must take this quiz in our classroom, and the only tools you are permitted to use are pencils, pens, erasers, and your mind. You have 1 hour. When you finish, please hand in your quiz to Sunny in his office (Tutt Science 234). Good luck!

Problem 1 (True-false, 10 points)**.** You will get 1 point for a correct answer and 0 points for blank and incorrect answers.[1] No explanations are required.

(1) There is a unique value of k which makes the vector $(k, 1, 1)$ orthogonal to $(3, 1, 4)$ in $\mathbb{R}^3$. **T** F

(2) There are infinitely many values of k such that the system T **F**

$$\begin{aligned} x + y &= 1 \\ 2x + 2y &= k \end{aligned}$$

has infinitely many solutions.

(3) A linear system with more variables than equations cannot have exactly one solution. **T** F

(4) The vector $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is in the set $\left\{ x \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + y \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix} : x, y \in \mathbb{R} \right\}$. T **F**

(5) There is a unique function $f : \mathbb{R} \to \mathbb{R}$ of the form $f(x) = ax^2 + bx + c$ whose graph passes through the three points $(1, 2), (-1, 6), (0, 3)$. **T** F

(6) The set T **F**

$$\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 1 \right\}$$

is a vector space (under the usual operations inherited from $\mathbb{R}^3$.)

(7) The set of diagonal 2×2 matrices **T** F

$$\left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \in \mathcal{M}_{2\times 2} : a, b \in \mathbb{R} \right\}$$

is a vector space (under the usual operations inherited from $\mathcal{M}_{2\times 2}$).

(8) The set of polynomials of degree at least 3, together with the zero polynomial, is a vector space (under the usual operations inherited from $\mathcal{P}$). T **F**

(9) The set of all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that $\int_0^1 f(x)\,dx = 0$ is a vector space (under the usual operations inherited from the vector space of all continuous functions). **T** F

(10) If $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$ are vectors such that $\vec{u}$ is orthogonal to $\vec{v}$ and $\vec{v}$ is orthogonal to $\vec{w}$, then $\vec{u}$ must be orthogonal to $\vec{w}$. T **F**

[1] During quiz revisions, you'll have the chance to meet with me one-on-one and convince me that you understand and can fully explain up to 2 of the true-false questions *that you left blank* to get full credit on those questions.

Problem 2 (3 points)**.** Solve the following system of equations. Show all of your work, and express the solution set in parametric form using vectors. Also, write a sentence explaining what the solution set looks like geometrically.

$$\begin{aligned} 2x - y - z &= 4 \\ x + y + z &= -1 \end{aligned}$$

Solution. We set up the augmented matrix

$$\left(\begin{array}{ccc|c} 2 & -1 & -1 & 4 \\ 1 & 1 & 1 & -1 \end{array}\right)$$

and then we row reduce until we get

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -2 \end{array}\right).$$

Thus the solutions (x, y, z) satisfy

$$x = 1 \text{ and } y + z = -2.$$

In other words, solutions are vectors of the form

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -2 - z \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Geometrically, the solution set is the line passing through the point $(1, -2, 0)$ in the direction of the vector $(0, -1, 1)$.

Problem 3 (3 points)**.** Sunny tells you that he has a piggy bank holding pennies, nickels and dimes that contains 12 coins with a total value of 92 cents. Is this possible? If so, how many coins of each type are in the box? If not, why not? Explain your answer fully.

Solution. Let x, y, z be the numbers of pennies, nickels, and dimes, respectively. The problem tells us that

$$x + y + z = 12$$
$$x + 5y + 10z = 92.$$

Row reducing to solve this system, we find that z is a free variable and

$$x = -8 + \frac{5}{4}z \text{ and } y = 20 - \frac{9}{4}z.$$

We want to find nonnegative integers x, y, z which satisfy these equations.

Note that z must be between 0 and 9. One option is that we could test all the possible integer values of z from 0 to 9 and find the value(s) of z that makes x and y also be nonnegative integers, if there is any such value of z. It will turn out that $z = 8$ is the only such value.

Alternatively, we can be lazy and use a bit of mathematical reasoning so that we have to do fewer calculations. In order for x and y to also be integers, we see that z must be divisible by 4 (otherwise the fractions won't go away). In order for x to be nonnegative, we need $z \geq 32/5 = 6.4$. Thus the only possibility is $z = 8$. Plugging into the above equations, we find $x = 2$ and $y = 2$, which are in fact both nonnegative integers, so this is a valid solution.

So it is possible for Sunny to be telling the truth; and, if he is telling the truth, his piggy bank must have 2 pennies, 2 nickels, and 8 dimes.

Problem 4 (Challenge problem! 3 points)**.** Is it true that any vector in $\mathbb{R}^2$ can be written in the form

$$x\begin{pmatrix}1\\3\end{pmatrix}+\vec{v}$$

for some $x \in \mathbb{R}$ and some $\vec{v} \in \mathbb{R}^2$ that is orthogonal to $\begin{pmatrix}1\\3\end{pmatrix}$? If so, prove it. If not, explain why not.

Solution. Note that vectors that are orthogonal to $(1,3)$ are all of the form $(-3y, y)$ for some $y \in \mathbb{R}$. So for any $(a,b) \in \mathbb{R}^2$ we would like to find x and y such that

$$x\begin{pmatrix}1\\3\end{pmatrix}+y\begin{pmatrix}-3\\1\end{pmatrix}=\begin{pmatrix}a\\b\end{pmatrix}.$$

This corresponds to the following linear system of equations.

$$\begin{aligned}x-3y&=a\\3x+y&=b\end{aligned}$$

So we set up the augmented matrix and row reduce.

$$\left(\begin{array}{cc|c}1&-3&a\\3&1&b\end{array}\right)\to\left(\begin{array}{cc|c}1&-3&a\\0&10&b-3a\end{array}\right)$$

There's a leading entry in every non-augmented column here, so this system must have a solution.

Honor code. If you have neither given nor received any unauthorized aid on this quiz, please write either "HCU" or "Honor Code Upheld" below, and sign your name next to it.

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